DIFFERENTIAL EQUATIONS - MATH 251

DIFFERENTIAL EQUATIONS

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DEFINITION

1 DEFINITION

A differential equation is an equation with an unknown function and one or more of its derivative.

A differential equation is an equation relating some function f to one or more of its derivatives.

Example:

$$\frac{d^2f}{dx^2} + 2\frac{df}{dx} + f^2x = sinx \tag{1}$$



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CLASSIFICATION

2 CLASSIFICATION

Differential equation can be classified by the following:

- Type: The equation of derivatives are taken with respect to one or several variables. Example

$$\frac{dx}{dy} = yx(ODE) \tag{2}$$

$$\frac{\delta y}{\delta t} + \frac{\delta y}{\delta x} = 0(PDE) \tag{3}$$

- Order: The order of a DE is the **order** of the highest derivative in that particular equation.



Examples.

$$(\frac{dy}{dx})^2 + \frac{d^2y}{dx^2} = sinx(order2) \tag{4}$$

$$\frac{d^3x}{dy^3} + x^{ll} + tx^2 = yx(order3) \tag{5}$$



Linearity: A DE is said to be linear if the following conditions are satisfied;

- the dependent variable(s) and their derivatives appear linear or in the power of one.
- There is no product of the dependent variable and any of its derivatives.
- There is no transcendental function (Trignometric, Logarithmic or exponential) of the dependent variable).



Examples

$$\frac{d^4x}{dt^4} = 0(linear) \tag{6}$$

$$x^{ll} + tx^2 = t(non - linear) \tag{7}$$

- Homogeneity: if the equation is not equal to a function of the independent variable. Example

$$\frac{dy}{dx}^2 + \frac{d^2y}{dx^2} = 0(homogeneous) \tag{8}$$

$$\frac{d^3x}{dy^3} + x^{ll} = sinx(non - homogeneous) \tag{9}$$



- Degree: The degree is the exponent of the highest derivative.

  Example

$$\left(\frac{dy}{dx}\right)^2 + \frac{d^2y}{dx^2} = sinx(degree1) \tag{10}$$



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SOLUTION TO A DIFFERENTIAL EQUATION

A function that satisfies the given differential equation is called its solution.

- The solution that contains as many arbitrary constants as the order of the differential equation is called a general solution.
- The solution free from arbitrary constants is called a particular solution.

Examples

Show that $y(x) = x^{\frac{-3}{2}}$ is a solution to the differential equation $4x^2y'' + 12xy' + 3y = 0$ for $x > 0$

solution

$y' = \frac{-3}{2}x^{\frac{-5}{2}}$

$y'' = \frac{15}{4}x^{\frac{-7}{2}}$

substituting this into the question yields

$4x^2(\frac{15}{4}x^{\frac{-7}{2}}) + 12x(\frac{-3}{2}x^{\frac{-5}{2}}) + 3(x^{\frac{-3}{2}}) = 0$

$0 = 0$

EXAMPLE

Find the DE whose general equation is

$y = c_1e^{-2x} + c_2e^{3x}$



SOLUTION TO A DIFFERENTIAL EQUATION

There are two forms of solution to a differential equation. They are:

- General Solution
- Particular Solution

General Solution

A general solution is a function that gives all the possible solutions to a differential equation.



Example: Solve the ODE $\frac{dy}{dx} = -4$

Solution

$\int dy = \int -4dx$

$y = -4x + c$ where c is an arbitrary constant.

This type of solution is a general solution.

Particular Solution

A Particular Solution is a solution of a differential equation taken from the General Solution by allocating specific values to the random constants.

Example: Solve $y' = -4, y(0) = 1$
from the general solution we have
$y = -4x + c$
substituting the condition given yields
$1 = 4(0) + c$
$c = 1$
Hence the particular solution becomes
$y = -4x + 1$

SUBSIDIARY CONDITION

A subsidiary condition helps to find a particular solution to a differential equation.

There are two types of subsidiary conditions.

- Initial conditions
- Boundary conditions

Example:

1. $y' = -4, y(0) = 1$ (Initial condition)
2. $y'' = -4, y(0) = 1, y'(0) = 2$ (Initial condition)
3. $y'' + 2y' = e^2, y(0) = 1, y'(1) = 1$ (boundary (0,1) condition)

NB: Both boundary and initial conditions gives a particular solution.

EXAMPLE

Given that $y = c_1e^{-x} + c_2e^{3x}$ is a general solution of the D.E
$y'' - 2y' - 3y = 0, y(0) = 3, y'(0) = 4$. Find the particular solution.



SOLUTION

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$$y = c_1 e^{-x} + c_2 e^{3x}$$

$$3 = c_1 + c_2$$

$$y' = -c_1 e^{-x} + 3c_2 e^{3x}$$

$$4 = -c_1 + 3c_2$$

Solving simultaneously yields

$c_1 = \frac{5}{4}$ and $c_2 = \frac{7}{4}$

Hence the particular solution becomes

$$y = \frac{5}{4} e^{-x} + \frac{7}{4} e^{3x}$$



First order ODEs are in the form $F(x, y, y') = 0$ There two forms ODEs. They are

- Standard Form

  $y' = f(x, y)$

  $\frac{dy}{dx} = f(x, y)$

- Differential form.

  $M(x, y)dx + N(x, y)dy = 0$

  $(x + y)dx + y^2 dy = 0$



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METHODS OF SOLVING ODEs

4 METHODS OF SOLVING DIFFERENTIAL EQUATIONS

- Separation of variables
- Homogeneous Method
- Integrating factor
- Exactness



METHOD OF SEPARATION OF VARIATION

4 METHODS OF SOLVING DIFFERENTIAL EQUATIONS

$M(y)\frac{dy}{dx} = N(x)$

$M(x,y)dy = N(x,y)dx$

let M(x,y)=m(x).p(y)

N(x,y)=n(x).q(y)

m(x).p(y)dx + n(x).q(y)dy=o

$\frac{m(x)}{n(x)} \cdot \frac{p(y)}{p(y)}dx + \frac{n(x)}{n(x)} \cdot \frac{q(y)}{p(y)}dy = 0$

$\implies \int \frac{m(x)}{n(x)}dx + \int \frac{q(y)}{p(y)}dy = c$



Example

Solve the ODE
$(x+1)\frac{dy}{dx} = x(y^2+1)$



Solution

4 METHODS OF SOLVING DIFFERENTIAL EQUATIONS

$$(x+1)dy = x(y^2+1)dx$$
$$(x+1)dy = (xy^2 + xdx)$$
$$(x+1)dy - (xy^2 + xdx) = 0$$
$$\frac{x}{x+1}dx - \frac{1}{y^2+1}dy = 0$$
$$\int \frac{x}{x+1}dx - \int \frac{1}{y^2+1}dy = C$$
let $u = x + 1$
$$x = u - 1$$
$$\int \frac{u+1}{u}du - \int \frac{1}{y^2+1}dy = C$$
$$\int udu \int \frac{1}{u}du - \int \frac{1}{y^2+1}dy = C$$
$$(x+1) - ln|x+1| - tan^{-1}y = C$$

TRIAL

Solve the following ODEs

1. $\frac{dy}{dx} = \frac{x^2}{y(1+x^3)}$
2. $(1+x^2) - x(1+2y)y' = 0$
3. $y' = \frac{3x^2+4x-4}{2y-4}, y(1) = 3$

Definition

A function f(x,y) is homogeneous if $f(\lambda x, \lambda y)=\lambda^n f(x,y), n = -\infty, ., ., .\infty$

Example: Check if $f(x,y) = x^4 - x^3y$ is homogeneous.

Solution

$$f(\lambda x, \lambda y) = (\lambda x)^4 - (\lambda x)^3(\lambda y)$$

$$f(\lambda x, \lambda y) = \lambda^4 x^4 - \lambda^3 x^3 \ \lambda y$$

$$f(\lambda x, \lambda y) = \lambda^4 x^4 - \lambda^4 x^3 y$$

$$\implies f(\lambda x, \lambda y) = \lambda^4 (x^4 - x^3 y)$$

Hence homogeneous.

Trials

Check for homogeneity in the following function

- $f(x,y) = \frac{2x+y}{x^2y^2}$
- $f(x,y) = e^{\frac{y}{x}} + tan(\frac{y}{x})$



HOMOGENEOUS METHOD

4 METHODS OF SOLVING DIFFERENTIAL EQUATIONS

In using the homogeneous method, we follow the following rules:

- Check for homogeneity of the function.
- Divide through the function by the highest term of the independent variable.
- Let $v = \frac{y}{x}$

EXAMPLES

Determine whether the differential equation is homogeneous, hence solve it.
$(x^2 + y^2)dx - 2xydy = 0$

Solution

$\frac{dy}{dx} = \frac{(x^2+y^2)}{2xy}$

From the funciton on the right hand side, we could see that, each term is of the power two(2). Hence, the function is homogeneous.

Let $v = \frac{y}{x}$

$\frac{dy}{dx} = v + x\frac{dv}{dx}$

$dy = vdx + xdv$

$$(x^2 + (xv)^2)dx - 2x(xv)(xdv + vdx) = 0$$

$$x^2dx + x^2v^2dx - 2x^3vdv - 2x^2v^2dx = 0$$

$$\frac{x^2dx}{x^2} - \frac{2x^3v}{x^2} - \frac{x^2v^2}{x^2} = 0$$

$$dx - v^2 - 2xvdv = 0$$

$$\frac{(1-v^2)}{1-v^2}dx - \frac{2xv}{x}dv = 0$$

$$\int \frac{1}{x}dx - \int \frac{2v}{1-v^2}dv = 0$$

$$\ln|x| = \ln|C| - \ln|1 - v^2|$$

$$\ln|x| = \ln\frac{C}{1-v^2}$$

$x = \frac{C}{1-v^2}$

but $v = \frac{y}{x}$

$x = \frac{C}{1-(\frac{y}{x})^2}$

$x = \frac{x^2 C}{x^2-y^2}$

$y = \sqrt{x^2 - Cx}$

TRIAL

1. $\frac{dy}{dx} = \frac{x^2+y^2}{xy}$
2. $2xdy = (x+y)dx$
3. $\frac{dy}{dx} = \frac{y^2-x^3}{yx}$
4. $\frac{dy}{dx} = \frac{2x+y}{x^2y^2}$



HOMOGENEOUS LINEAR COEFFICIENT FUNCTIONS

Given an ODE in the form $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$,
we say this type of ODE is almost homogeneous. There are two ways of solving the problem.

Case 1:

When $\frac{a_2}{a_1} \neq \frac{b_2}{b_1}$

Case 2:

When $\frac{a_2}{a_1} = \frac{b_2}{b_1}$



CASE 1

4 METHODS OF SOLVING DIFFERENTIAL EQUATIONS

Given an ODE in the form $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$

When $\frac{a_2}{a_1} \neq \frac{b_2}{b_1}$

Set $x = X + h, y = Y + k$

where $(a_1X + b_1Y)dx + (a_2X + b_2Y)dy = 0$

$(a_1h + b_1k + c_1) = 0$

$(a_2h + b_2k + c_2) = 0$

Then we solve for h and k.



Example

Solve the first order ODE $(x - 2y + 1)dx + (4x - 3y - 6)dy = 0$

Solution

$\frac{a_2}{a_1} = \frac{4}{1} \neq \frac{b_2}{b_1} = \frac{3}{2}$

Set $x = X + h, y = Y + k$

$h - 2k + 1 = 0$

$4h - 3k - 6 = 0$

Solving this simultaneous $h = 3, k = 2$

$(X - 2Y)dX + (4X - 3Y)dY = 0$

$$\frac{dY}{dX} = -\frac{X-2Y}{4X-3Y}$$

$$= \frac{1-2\frac{Y}{X}}{3\frac{Y}{X}-4}$$

$$vdX + xdv = \frac{1-2v}{3v-4}dX$$

$$\frac{3v^2-2v-1}{3v-4}dX = -Xdv$$

$$-\int \frac{1}{X}dX = \int \frac{3v-4}{3v^2-2v-1}$$

$$-\ln|X| = \frac{15}{4}\ln|3v+1| - \frac{1}{4}\ln|v-1| + \ln|c|$$

$$-4\ln|X| = 15\ln|3v+1| - ln|v-1| + 4\ln|c|$$

$$\ln|X^{-4}c^4| = \ln|\frac{(3v+1)^{15}}{v-1}|$$



$$X^{-4}c^4 = \frac{(3v+1)^{15}}{v-1}$$

$$X^{-4}c^4 = \frac{((\frac{Y}{X})+1)^{15}}{(\frac{Y}{X})-1}$$

$$X^{-5}c^4 = (\frac{3Y+x}{X})^{15}/(Y-X)$$

$$X^{10}c^4 = \frac{(3Y+X)^{15}}{Y-X}$$

$$X^{10}c^4 = (3Y+X)^{15}$$

$$(x-3)^{10}c^4Y - X = (3y-6+x-3-9)^{15}$$

$$(x-3)^{10}(y+x-5)c^4 = (3y+x-9)^{15}$$

Trials

1. $(2x + y - 2)dx + (-x + 2y + 1)dy = 0$
2. $(x - 3y)dx + (x + y - 4)dy = 0$



case 2

Given an ODE in the form $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$

When $\frac{a_2}{a_1} = \frac{b_2}{b_1} = K$
Set $z = a_1x + b_1y$

Example:

Solve the ODE $(x + 2y + 3)dx + (2x + 4y - 1)dy = 0$

Solution

$\frac{a_2}{a_1} = \frac{2}{1} = \frac{b_2}{b_1} = \frac{4}{2} = 2$

let $z = x + 2y$

$\frac{dz}{dx} = 1 + 2\frac{dy}{dx}$

$\frac{dz-dx}{2} = dy$

$(z+3)dx + (2z-1)(\frac{dz-dx}{2}) = 0$

$2(z+3)dx + (2z-1)(dz-dx) = 0$

$(2z+6)dx + 2zdz - 2zdx - dz + dx = 0$

$$\int 7dx + \int (2z - 1)dz = 0$$

$$7x + z^2 - z = c$$

$$7x + (x + 2y)^2 - (x + 2y) = c$$

Trial

1. $(x + 2y + 1)dx + (2x + 4y + 3)dy = 0$
2. $(x - y)dx + (x - y + 2)dy = 0$



INTEGRATING FACTOR

4 METHODS OF SOLVING DIFFERENTIAL EQUATIONS

In order to solve a linear first order differential equation we MUST start with the differential equation in the form $\frac{dy}{dt} + p(t)y = g(t)$ where both p(t) and g(t) are continuous functions.

We find the solution to such a problem by multiplying a function called the **integrating factor** $\mu(t) = e^{\int p(t)dt}$ through the DE. i.e. $\mu(t)\frac{dy}{dt} + \mu(t)p(t)y = \mu(t)g(t)$.

This reduces to $\frac{d}{dt}\mu(t)y = \mu(t)g(t)$ (A derivative of the product of the integrating factor and the dependent variable).



Example

Solve the differential equation
$\frac{dy}{dt} - 2y = 4 - t$

Solution

$\mu(t) = e^{\int -2dt}.$

$\mu(t) = e^{-2t}$

$e^{-2t}\frac{dy}{dt} = 4e^{-2t} - te^{-2t}$

$e^{-2t}dy = (4e^{-2t} - te^{-2t})dt$

$\int e^{-2t}dy = \int (4e^{-2t} - te^{-2t})dt$

$e^{-2t}y = -2e^{-2t} - (\frac{-t}{t}e^{-2t} - \frac{1}{4}e^{-2t}) + C$

$e^{-2t}y = -2e^{-2t} + \frac{-t}{t}e^{-2t} - \frac{1}{4}e^{-2t} + C$

$y = \frac{t}{2} - \frac{7}{4} + ce^{2t}$

Exercise

Solve the following IVP differential equations

- $\frac{dv}{dt} = 9.8 - 0.196v, v(0) = 48$
- $ty' + 2y = t^2 - t + 1, y(1) = \frac{1}{2}$
- $ty' - 2y = t^5 sin(2t) - t^3 + 4t^4, y(\pi) = \frac{3}{2}\pi^4$

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Thank you for listening!
Any questions?